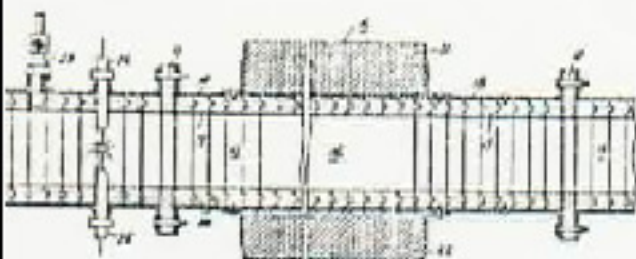


flow passage being closed at both ends, magnetic means about said annular flow passage for imposing a steady radial magnetic field across said flow passage, means associated with said flow passage for imposing the conducting fluid to flow alternately in opposite directions through said magnetic field to establish a circulating current within the moving fluid by interaction of said field and the fluid, and coil means connected to have an alternating electromotive force induced therein by the varying circulating current in the

3,149,251

ELECTRIC POWER GENERATION

Ernest Hurwitz, Jr., Schenectady, N.Y., assignor to General Electric Company, a corporation of New York
Filed Apr. 17, 1961, Ser. No. 103,303
8 Claims. (Cl. 310-11)

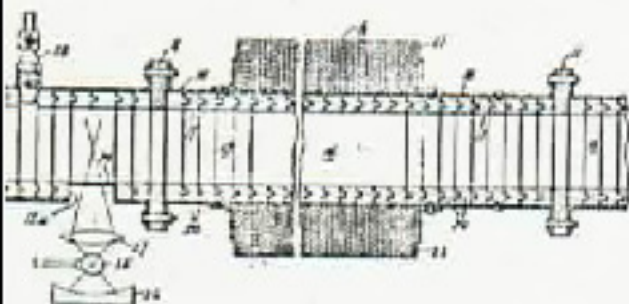


A method of generating electric power, which comprises the steps of injecting an easily ionizable first material into a gaseous fluid, injecting into said gaseous fluid a second material whose atoms have at least one metastable energy level, raising the energy level of atoms of said first material to their metastable level whereby said first material are ionized by collisions with atoms of said second material in their metastable level, passing the gaseous fluid mixture through a magnetic field whereby electric current flows through said gaseous fluid.

3,149,252

ELECTRIC POWER GENERATION

William Lapp, Schenectady, N.Y., assignor to General Electric Company, a corporation of New York
Filed Apr. 17, 1961, Ser. No. 118,212
4 Claims. (Cl. 310-11)



In a magnetohydrodynamic generator, the combination comprising means defining a flow path for a gaseous fluid, means for injecting into the gaseous fluid an ionizable first material and a second material whose atoms have at least one metastable energy level comparable to the ionization energy level of the first material, irradiation means for raising the energy level of the atoms of said second material to a level above their metastable level, means for causing the atoms to drop to their metastable level whereby atoms of the first material are thereionized by collisions with atoms of the second material to their metastable level.

ELECTRODE STRUCTURE FROM MAGNETO-HYDRODYNAMIC DEVICE

Emmett A. Luebke, Schenectady, N.Y., assignor to General Electric Company, a corporation of New York
Filed Jan. 3, 1962, Ser. No. 164,114
7 Claims. (Cl. 310-11)



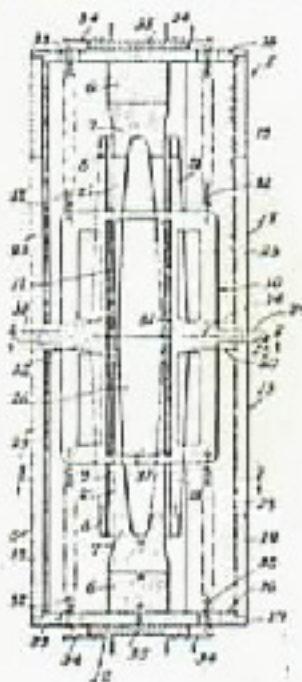
1. An electrode for a magnetohydrodynamic device which comprises a bonded mass of refractory thermionic emitting particles having a conducting network of refractory metal filaments dispersed throughout the mass.

3,149,254

LINEAR MOTOR OR GENERATOR

Thomas A. Carter, 7914 Melita Ave., North Hollywood, Calif., and Glen D. Sumerlin, 13606 Kismet, Sylmar, Calif.

Filed Aug. 7, 1961, Ser. No. 129,831
8 Claims. (Cl. 310-12)



1. A linear motor-generator comprising a permanent magnet, an inner pole piece extending from one pole of said magnet, a magnetic-transfer housing connected to and partly enclosing said magnet and inner pole piece, an outer pole piece carried by and within the magnetic-transfer housing around and partly coextensive with the inner pole piece, the inner pole piece being of similar polarity to the pole of the magnet from which it extends, the outer pole piece being of opposite polarity to the inner pole piece, said two pole pieces defining an elongated and annular flux cavity therebetween, and a free, movable coil in said cavity.

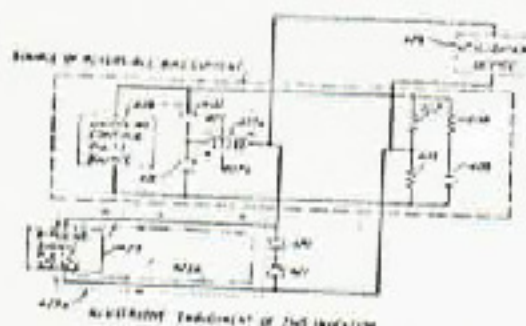
3,149,255

ELECTRICAL RECIPROCATING MOTOR

Anthony B. Trench, Galt, Ontario, Canada, assignor to H & T Electrical Products, Toronto, Ontario, Canada
Filed Mar. 23, 1962, Ser. No. 183,326
20 Claims. (Cl. 310-30)

1. An electromagnetic axially reciprocating motor comprising a generally tubular stator having laterally

ing an inductor, means for controlling the direction of current flow through said inductor, and a circuit configura-



tion including two series-opposed negative resistance diodes connected in series with said inductor.

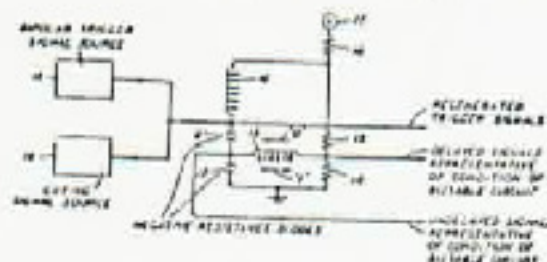
3,116,425

BISTABLE STAGES HAVING NEGATIVE RESISTANCE DIODES AND INDUCTORS

Reginald A. Kaenel, Murray Hill, N.J., assignor to Bell Telephone Laboratories, Incorporated, New York, N.Y., a corporation of New York

Filed June 27, 1960, Ser. No. 39,117

13 Claims. (Cl. 307-88.5)



2. In combination, first circuit path means comprising voltage-controlled negative resistance diodes connected in series-aiding and first inductance means connected in series with said diodes, second circuit path means connected in parallel with said first circuit path means and comprising two series-connected impedance elements, second inductance means connected between the midpoint of said series-aiding diodes and the midpoint of said series-connected impedance elements, source means for biasing one of said diodes to the relatively low voltage positive resistance region of its characteristic curve and for biasing the other one of said diodes to the relatively high voltage positive resistance region of its characteristic curve, and signal source means connected to the junction of said first inductance means and said series-aiding diodes for applying a signal to said diodes to cause them to interchange their positions on said positive resistance regions and thereby to cause a regenerated signal to appear across said diodes.

3,116,426

LOGIC CIRCUITS EMPLOYING BRIDGE NETWORKS COMPRISING TRANSFORMER SECONDARIES AND N-TYPE CONDUCTIVITY CURVE NEGATIVE RESISTANCE ELEMENTS

Shintaro Oshima, Musashino-shi, Hajime Enomoto, Ichikawa-shi, Shlyoji Watanabe, Shinjuku-ku, Tokyo-to, and Katsuro Amano, Ota-ku, Tokyo-to, Japan, assignors to Kokusai Denshin Denwa Kabushiki Kaisha, Tokyo-to, Japan, a joint-stock company of Japan

Filed Nov. 14, 1960, Ser. No. 69,165

Claims priority, application Japan Nov. 16, 1959

14 Claims. (Cl. 307-88.5)

13. A logical element comprising, a bridge circuit comprising a series combination of elements having n-type conductivity connected in the same sense in said bridge circuit, at least one transformer secondary winding connected forming two arms of said bridge circuit, means

to establish alternatively two stable states of operation in said n-type elements corresponding to positive and negative polarities of an information signal upon application of said information signal comprising means for applying said information signal to said bridge, an output circuit connected to said n-type elements, connections



connected to a connection point corresponding to the electrical center of said secondary winding for applying bias voltages thereto simultaneously with the application of said information signal to develop at said output circuit an output signal having a polarity corresponding to the polarity of said information signal.

3,116,427

ELECTRON TUNNEL EMISSION DEVICE UTILIZING AN INSULATOR BETWEEN TWO CONDUCTORS EITHER OR BOTH OF WHICH MAY BE SUPERCONDUCTIVE

Ivar Giaever, Schenectady, N.Y., assignor to General Electric Company, a corporation of New York

Filed Nov. 17, 1960, Ser. No. 70,074

11 Claims. (Cl. 307-88.5)



1. An electronic device of the type described comprising a first conductor, an insulator, and a second conductor in a sandwich-like arrangement with the insulator in the form of an electron-permeable continuous film of thickness between two and 1000 angstroms disposed between and separating the first and second conductors and contacting opposed surfaces of the said conductors, the first conductor having a normal resistance state and below a critical temperature having a superconductive state and means for maintaining said first conductor at a temperature below its superconducting critical temperature.

3,116,428

FORCE RESPONSIVE ELECTRICAL IMPULSE GENERATOR

William G. Blodgett, Redondo Beach, and Francis M. Cummings and Walter H. Fleming, Los Angeles, Calif., assignors to Space Equipment Corporation, Torrance, Calif., a corporation of California

Filed Oct. 16, 1961, Ser. No. 145,369

1 Claim. (Cl. 310-15)

An impulse generator comprising, in combination: a cylindrical container; a stationary magnetic member extending transversely of said container and rigidly secured to the sidewalls of the container; electrical windings secured to said stationary magnetic member; a movable magnetic member having a general cup-shaped configuration with a central upwardly extending core, one of said magnetic members being permanently magnetized, said movable magnetic member receiving said windings within its sidewalls and over its core in flux coupling relationship

tors being substantially parallel to a plane passed through said second array of parallel conductors; an electroluminescent layer interposed between said arrays; and a plurality of discrete spaced pillars interposed between one of



said arrays and said electroluminescent layer, each of said pillars being composed of a symmetrically non-linear resistive core circumferentially surrounded by an electrically insulating coating.

3,184,636

COLD CATHODE

Burnell V. Dore, Sunnyvale, and Richard A. Hein, Palo Alto, Calif., assignors to Sylvania Electric Products Inc., a corporation of Delaware

Filed June 15, 1961, Ser. No. 117,236

8 Claims. (Cl. 315-94)



1. An electron emission device comprising a metallic substrate, a barrier layer of semiconductor material formed on said substrate, an emission layer formed on said barrier layer, said emission layer comprising a combination of a conductor having a relatively low work function and an oxide of an alkali earth metal, a source of bias voltage, and means for connecting said source to said substrate and to said conductor for applying an electric field across the barrier layer.

3,184,637

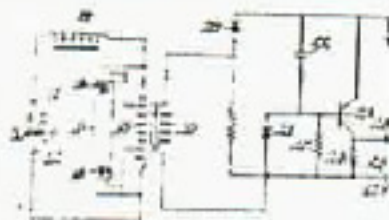
LAMP MONITORING APPARATUS

Peter Skinner, London, England, assignor to Decca Limited, London, England, a British company

Filed Dec. 14, 1962, Ser. No. 243,230

Claims priority, application Great Britain, Dec. 13, 1961, 44,545/61

5 Claims. (Cl. 315-129)



1. In a fluorescent lamp system having a fluorescent lamp with a transformer winding connected in shunt across the lamp and a ballast choke in series with the lamp in an alternating supply circuit, monitoring apparatus comprising an auxiliary winding inductively coupled to the lamp circuit to have an induced alternating signal of magnitude dependent on the relative potential distribution across the ballast choke and transformer winding, a rectifier arranged to produce a direct control signal of magnitude dependent on the magnitude of said induced

alternating signal, an auxiliary supply source, an output circuit, and switch means controlled by said control signal and controlling the application of potential from said auxiliary supply source to said output circuit, said switch means being arranged to reduce the potential in the output circuit unless the control voltage corresponds to a potential distribution in the lamp circuit with substantially the full alternating supply voltage across the transformer.

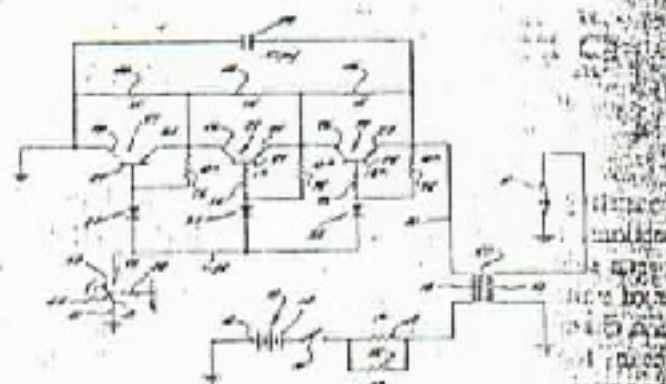
3,184,638

SERIES STACKED TRANSISTORIZED IGNITION SYSTEM HAVING DIODES IN SERIES WITH THE BASES

Warren R. Wood, Palo Alto, Calif., assignor to Robert Bosch G.m.b.H., Stuttgart, Germany

Filed May 10, 1962, Ser. No. 193,765

4 Claims. (Cl. 315-206)



1. An improved ignition system for an internal combustion engine having spark plugs and distributor points for opening and closing a circuit in synchronization with engine operation, the system comprising a source of electrical power, an ignition transformer having a primary winding and a secondary winding, means connecting the secondary winding to the spark plugs, a plurality of transistors each having a base, emitter and collector and an emitter-base path, means connecting the transistors in series so the emitter of one is connected to the collector of an adjacent transistor, means connecting the transistors in series with the source of electrical power and the primary winding so the transistors control the flow of current through the primary winding, a separate biasing resistor connected across the base and emitter of each transistor, means connecting the biasing resistors in circuit with the source of power and distributor points to apply a pulsating voltage across the biasing resistors and control the flow of current through the transistors, a separate diode connected to and in series with the base of each transistor and the distributor points to isolate the transistors when the points open, and a separate dropping resistor connected in series with the base of each transistor between said base and the corresponding diode to limit base current flow, each of said biasing resistors and each of the corresponding ones of said dropping resistors being connected in series circuit with each other and with the corresponding emitter-base path to form a closed loop comprising a biasing resistor, a dropping resistor and the emitter-base path for each of said transistors.

3,184,639

SPARK IGNITION APPARATUS

Norman Arthur Parish, Hemel Hempstead, England, assignor to Rotax Limited, London, England

Filed Oct. 2, 1961, Ser. No. 142,123

Claims priority, application Great Britain, Oct. 11, 1960, 34,823/60

2 Claims. (Cl. 315-209)

1. Spark ignition apparatus comprising in combination first and second terminals for connection to a D.C. source, a transformer having a primary winding and a secondary

winding, a capacitor winding in a series gap and a spark said capacitor, winding to said which the other to the first terminal interrupter to a magnetic forces winding for urging against the tional winding ha

first and said second to current flow ing the action of the breakdown voltage operable when the is reduced for dis the first and second of said interrupter ing and said addi and disconnection maintain the spark breakdown voltage

INTERNAL C**SYSTEM**

Norman Alfred

Joseph Lucas

England

Filed in

Claims priority, a

An internal co prising in combin said generator inc

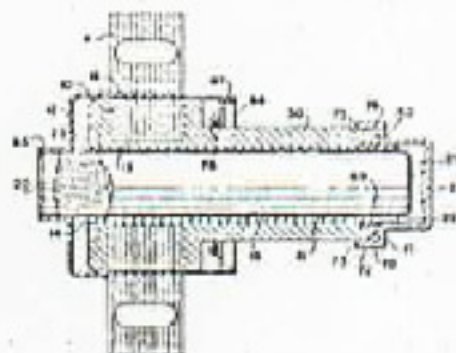
of uniform diameter terminating at a radial shoulder, an annular field core slidably mounted in and closely fitting said bore throughout the length of the core, the inner end of the latter being seated against said shoulder, said second member being in end-to-end abutment with said first member and having an axially extending pilot portion slidably extending into said bore against the other end of said core, and means for securing said members in assembled relation.

3,150,279

BEARING CONSTRUCTION

Elmer Fred Macks, Willow Lane, Vermilion, Ohio
Original application Jan. 14, 1959, Ser. No. 786,856.
Divided and this application June 14, 1962, Ser. No. 202,605

7 Claims. (Cl. 310-90)



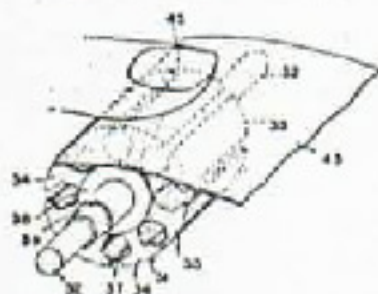
1. An article of manufacture including first and second relatively rotatable assemblies, said first assembly including a pin member, the second assembly including a sleeve support tube, a sleeve member supported within the tube and telescoped over the pin to define a complementary pneumatic load-carrying film producing region therebetween, an end cap secured to the tube and spaced from the pin to provide a trapped pocket of air therebetween, and an annular deformable ring squeezed between the cap and the tube and pressed against the sleeve to prevent relative axial movement of the sleeve.

3,150,280

INSULATED STACK OF ELECTRICAL LAMINATIONS AND METHOD OF MAKING THE SAME

Joseph R. Little, Springfield, Pa., assignor, by mesne assignments, to Litton Industries, Inc., Beverly Hills, Calif., a corporation of Delaware

Filed Oct. 13, 1960, Ser. No. 62,496
13 Claims. (Cl. 310-215)



1. An insulated rotor comprising a shaft having a stack of laminations mounted thereon, said laminations having a series of peripheral teeth with slots formed therebetween, a wire winding on said rotor positioned in said slots to form slot windings and positioned across the face of each end lamination to form end windings, and a single sheet of insulation lying between said windings and said stack, said insulation sheet lying in the slots to form slot insulation and across the face of each end lamination to form end insulation, said insulation sheet having rounded corners between said slot and end insulation, said end insulation being crinkled so as to cushion said end windings.

3,150,281

GLASS OF HIGH ULTRAVIOLET TRANSMITTANCE, METHOD, AND ARTICLES MANUFACTURED THEREFROM

Adli M. Bishay, Chicago, Ill., assignor to the United States of America as represented by the United States Atomic Energy Commission

Filed Apr. 26, 1961, Ser. No. 105,825

4 Claims. (Cl. 313-221)

1. A glass composition consisting essentially of the molar proportions $4.5\text{B}_2\text{O}_3$, $1.0\text{Al}_2\text{O}_3$, and $1.0\text{K}_2\text{O}$.
2. A method of making a glass of high ultraviolet transmittance, comprising melting under an inert atmosphere a mixture consisting essentially of the molar proportions $0.90(\text{NH}_4)_2\text{B}_{10}\text{O}_{16} \cdot 8\text{H}_2\text{O}$, $2.0\text{Al}(\text{OH})_3$, and $1.0\text{K}_2\text{CO}_3$.
3. A germicidal lamp comprising an elongated cylindrical envelope of glass consisting essentially of B_2O_3 , Al_2O_3 and K_2O in the molar proportions of $4.5\text{B}_2\text{O}_3$, $1.0\text{Al}_2\text{O}_3$ and $1.0\text{K}_2\text{O}$, said envelope being hermetically sealed at both ends around electrical conductors, a ballast at one of the ends, electrodes within the envelope at each end, and mercury vapor throughout the interior of the envelope.

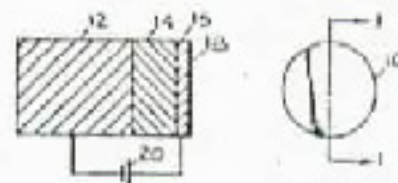
3,150,282

HIGH EFFICIENCY CATHODE STRUCTURE

Donovan V. Geppert, Menlo Park, Calif., assignor to Stanford Research Institute, Menlo Park, Calif., a corporation of California

Filed Nov. 13, 1963, Ser. No. 237,213

7 Claims. (Cl. 313-346)



1. A cathode structure comprising a first metal section, a contiguous large band-gap semi-conductor section, a second metal section contiguous to said semi-conductor section, and a surface layer for said second metal section, the metal of said first section being selected to have a low work function relative to the metal of said third section, said surface layer being selected to be a material which has a low work function relative to said first metal section.

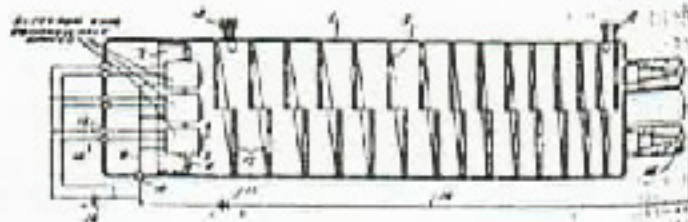
3,150,283

TRAVELING WAVE TUBE HAVING RECTANGULAR WAVEGUIDE SLOW WAVE STRUCTURE WITH EQUIANGULARLY SPACED ELECTRON GUNS

Joseph E. Rowe, Ann Arbor, Mich., assignor to the United States of America as represented by the Secretary of the Air Force

Filed May 19, 1961, Ser. No. 111,411

1 Claim. (Cl. 315-3.6)



A traveling wave tube comprising an evacuated metallic cylinder, a conductive helical ribbon having one edge in continuous electrical contact with the inner surface of said cylinder and having a width equal to the inner radius of said cylinder, said ribbon forming with the inner surface of said cylinder a helical waveguide slow wave structure, said ribbon having aligned apertures

each tube conducts on 1/2 of cycle

27 MHz

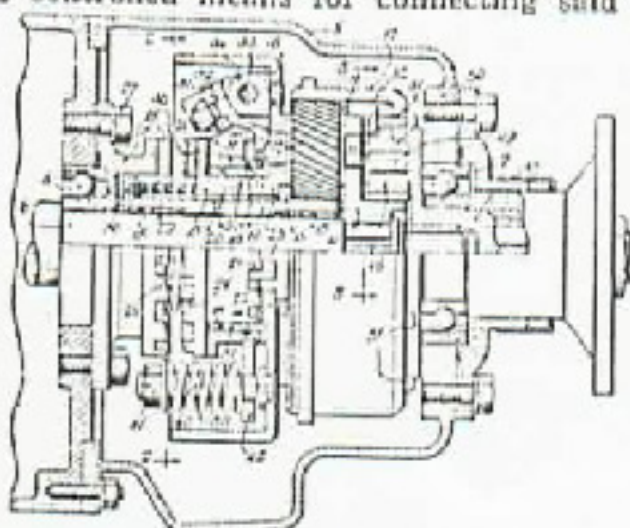
~~1 Kw~~

! ~200W output
tube filament
12V 10A 867

PHIL. 23, 1940

U. S. PATENT OFFICE

with said sun and orbit gears, and speed responsive controlled means for connecting said sun



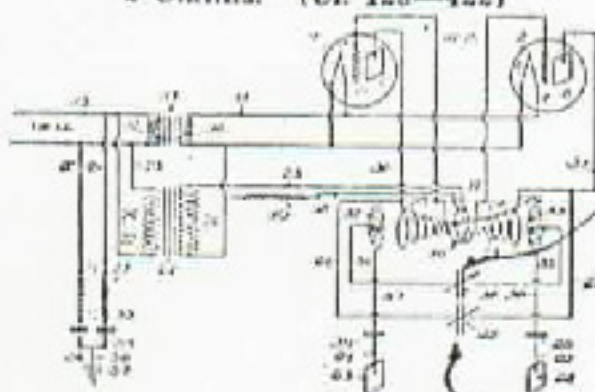
gear to revolve with said carrier or to hold it from rotation therewith.

2,198,073

SHORT WAVE THERAPY UNIT

Harold S. Bayless, Herman D. Bauer, and Hyman D. Fischer, Chicago, Ill.

Application September 28, 1936, Serial No. 102,912
2 Claims. (Cl. 128-422)



1. An electro-therapy device comprising thermionic valves having cathodes, grids and anodes, an input circuit, a transformed circuit from said input circuit to said cathodes to heat said cathodes, a transformed circuit from said input circuit to said grids and said anodes, a grid coil in the circuit of said grids, a conversion inductance coil in the circuit of said anodes, pick-up coils inductively coupled to said conversion coil, said coils being of tubular material to present inner and outer electrically active surfaces, said conversion and pick-up coils being so arranged and located as to be varied in angular relation of the axes thereof, said conversion coil having its convolutions so varied in diameter as to be in generally conoidal form, electrode pads electrically connected to said pick-up coils, and a variable capacity connected in shunt with said pick-up coils for tuning the resonance of the circuit to said electrode pads.

2,198,074

METHOD OF MAKING BEARINGS

Ernest V. Berry, Los Angeles, Calif.

Application February 6, 1939, Serial No. 254,755
2 Claims. (Cl. 29-149.5)



1. The method of producing a bearing sleeve
513 O. G.-57

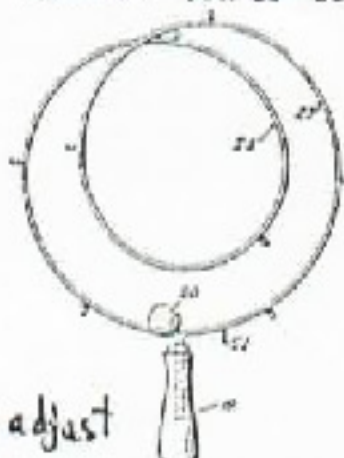
for a machine element which consists in utilizing the metal of a previously formed sleeve, reconstructing said sleeve by subjecting it to force in a machine so as to uniformly move the metal structure thereof into surfaces with new dimensions while substantially maintaining the original wall thickness, applying a coating of metal to one of the newly created wall surfaces and finishing said coating and the other newly created wall surface to fit the element journal and sleeve holder.

2,198,075

AMUSEMENT DEVICE

Benjamin Borek, Chicago, Ill.

Application March 10, 1937, Serial No. 130,032
2 Claims. (Cl. 46-51)



Power o/p adjust

1. In a device of the character described, a track comprising a pair of wires arranged to form a plurality of loops in planes forming an angle to each other, and means for maintaining said wires in spaced relation to afford a looped closed path for a ball.

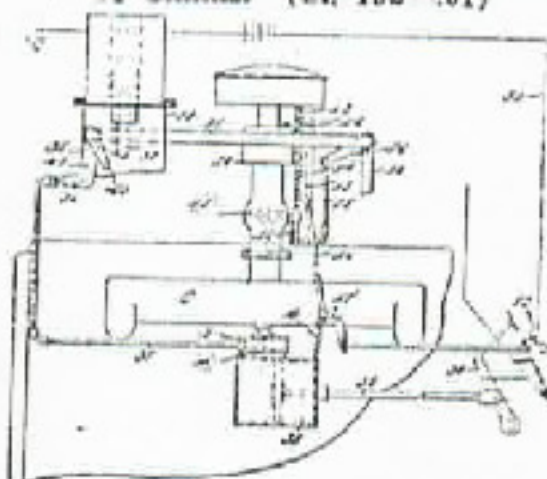
Resonant Freq. adjust for dip

2,198,076

MOTOR VEHICLE CONTROL

Forest S. Cartwright, Indianapolis, Ind.

Application February 6, 1937, Serial No. 124,449
14 Claims. (Cl. 192-01)



1. The combination of a motor of the internal-combustion type, a power train including a step-by-step speed-changing gearing, clutching means interposed between the motor and power train, a throttle valve controlling flow of fuel to the motor, operator-controlled means for positioning said throttle valve, power-actuated means for alternatively positioning said clutching means, a second power-actuated means for selectively positioning the speed changing gearing, including an operator-controlled selector, a control element for controlling the application of power to the power-actuated clutch-operating means, a third power-actuated means, connections between said third power-actuated means, the power-control-

being separated by relatively broad flat areas of said surface, terminal means at each end of said conductor for applying electrical current thereto, and a quantity of electrically insulative material formed in a spiral configura-

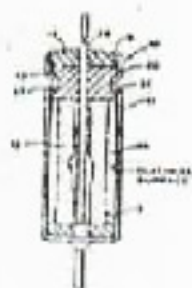


tion similar to that of said strip of conductor material and interlaminated therewith to form a unitary annular coil structure characterized by alternated turns of conductor and insulator and further characterized by transverse coolant passages formed by said grooves.

3,056,072

TANTALUM CAPACITOR

Walter W. Schroeder and David B. Peck, Williamstown, Mass., assignors to Sprague Electric Company, North Attacus, Mass., a corporation of Massachusetts
Filed June 26, 1958, Ser. No. 744,723
2 Claims. (Cl. 317-230)



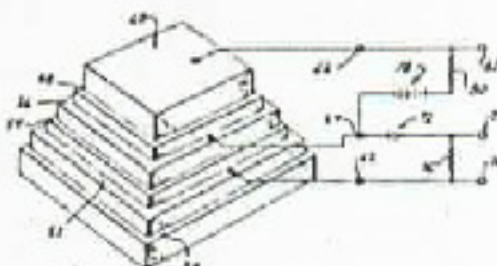
1. An electrolytic capacitor comprising a silver cathode container having an integral closed end, an inert spacer member within said container at said closed end, a pellet-type tantalum anode positioned on said spacer member, a corrosive electrolyte having an acid pH surrounding said anode in said container, a seal comprising a circumferentially grooved plug of essentially rigid halogenated ethylene polymer within said container external of said anode, a tacky hydrocarbon caulking material in said groove, said anode maintained between said plug and said spacer by the lip of said container being spun onto said seal, said capacitor sealed by the adjacent wall of said container being rolled into said groove whereby said caulking material maintains the seal upon differential expansion and contraction of said container and said plug.

3,056,073

SOLID-STATE ELECTRON DEVICES

Carver A. Mead, Pasadena, Calif., assignor to California Institute Research Foundation, Pasadena, Calif., a corporation of California

Filed Feb. 15, 1960, Ser. No. 8,589
13 Claims. (Cl. 317-234)



9. A solid-state electron device comprising a first and second conductive metal layer separated by a first layer of insulating material, said insulating layer of material

having a thickness less than the mean-free path of an electron, said second conductive metal layer having a thickness less than the mean-free path of an electron, a second layer of insulating material on said second conductive layer surface opposite to that adjacent said first layer of insulation material, said second insulating layer having a thickness less than that which will cause space charge limitation of the electrons emitted from said second conductive layer, and a third conductive metal layer on said second insulating material layer.

3,056,074

MINIATURE DESICCATOR AND PRODUCTION OF UNITS THEREOF

John Russell Fahey and George L. Schnable, Lansdale, Pa., assignors, by mesne assignments, to Philco Corporation, Philadelphia, Pa., a corporation of Delaware
Continuation of application Ser. No. 67,071, Nov. 3, 1960
This application Dec. 16, 1960, Ser. No. 91,292
15 Claims. (Cl. 317-234)



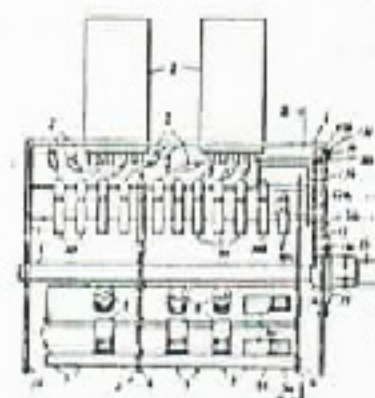
1. A miniature desiccator comprising a single granule of glass on a filament of metal, said granule comprising a porous silica lattice shell and a non-porous core of phase-separated borosilicate, said filament being firmly bonded to said core and extending through and from said shell.

3,056,075

TURRET TUNERS

Anthony S. Coleman and Hubert Roy Holland, Ascot, England, assignors to Brayhead (Ascot) Limited, Ascot, England, a British company

Filed Nov. 4, 1958, Ser. No. 771,759
Claims priority, application Great Britain Nov. 7, 1957
3 Claims. (Cl. 317-249)



1. A fine tuning arrangement for V.H.F. frequencies comprising a plurality of electrical condensers, said condensers consisting of a plurality of fixed plates side-by-side in the same plane and a common movable plate opposite said fixed plates, a supporting wall member arranged generally parallel to said plates and having a slot, said movable plate, which is located between said fixed plates and said supporting wall member, having an edge zone formed as a substantially Z-shaped portion engaging through said slot with the central part of the Z bearing against an edge of the slot, a finger on an opposite edge of said movable plate, a rotatable cam engaged by said finger, a spring member on the opposite side of said supporting wall member to said fixed and movable plates and bearing against the free end of said Z-shaped portion to urge said finger into engagement with said cam, where-

duce a varying signal serially applied to the voltages in said connections which signal amplitude varies at a frequency dependent upon the speed of said motor, and means to measure the difference in shaft rotation between said synchro transmitter and said synchro receiver.

2,960,656

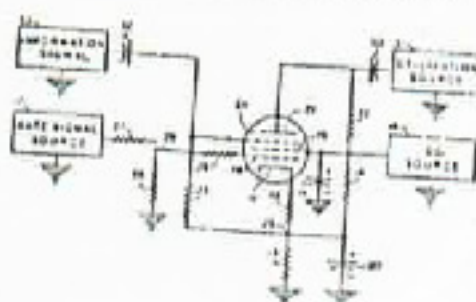
UNIPULSE GATE

Richard G. Cumings, 2716 Rumblewood Drive, District Heights, Md.

Filed June 30, 1959, Ser. No. 824,163

3 Claims. (Cl. 328-102)

(Granted under Title 35, U.S. Code (1952), sec. 266)



1. A unipulse gate comprising: a vacuum tube including at least a plate, a cathode, a center screen grid, a first control grid between said screen grid and said cathode, and a second control grid between said screen grid and said plate; a grounded positive D.C. potential source connected to said screen grid and said plate; a constant bias means connected to said cathode and said source for applying a potential to said cathode which is a minor fraction of the potential of said source; and self bias means connected between said second grid and said cathode for applying a potential to said second control grid less than the potential of said cathode by a factor directly proportional to the screen current.

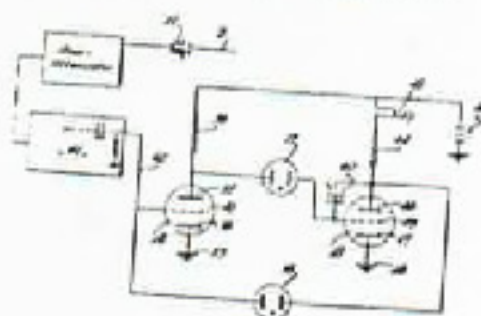
2,960,657

CIRCUIT MONITOR

Clement B. Edgerly, North Hollywood, Calif., assignor to Newcomb Electronics Corp., Los Angeles, Calif., a corporation of California

Filed Apr. 8, 1957, Ser. No. 651,527

6 Claims. (Cl. 328-200)



3. A circuit monitor for indicating the failure of an alternating current signal: a pair of variable electronic devices each having electrodes adapted to connect with a load circuit and also having a control electrode; means for forming load circuits for each of said devices; means for exciting said load circuit; an electronic discharge element connected between a point in the load circuit of one device and the control electrode of the other device, said electronic discharge element requiring the application of a predetermined potential for firing; means for applying the signal to be monitored to the control electrode of said one device; the load circuit for said one device being so characterized that the equilibrium potential of said point

discharge element firing in response to swings in one direction of the monitored signal; means for steadying the pulsations at the control electrode of the other device; and means operated in response to cessation of oscillation of said discharge element, and by the aid of the second device for imposing a bias on the control electrode of the first device to prevent the monitored signal, upon resumption, from causing said discharge element to fire; and reset means for restoring first electronic device for control by the monitored signal.

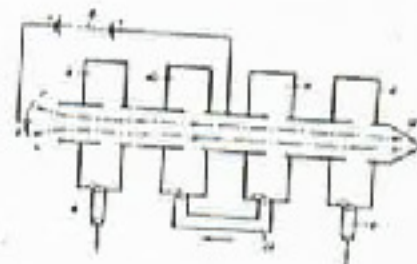
2,960,658

MICROWAVE AMPLIFIERS

John Dain, Chelmsford, and Percy Charles Ruggles, Great Baddow, England, assignors to English Electric Valve Company Limited, Kingsway, London, England, a British company

Filed Oct. 10, 1958, Ser. No. 766,514

Claims priority, application Great Britain Dec. 12, 1957
3 Claims. (Cl. 330-44)



1. A high gain microwave amplifier of the kind in which amplification is obtained by interaction between an electron beam and at least one resonator coupled thereto and adapted to modulate the velocity of the electrons thereof, said amplifier comprising an electron beam source, an input resonator coupled to said beam, means for applying input signals to said resonator to modulate said beam in accordance therewith, an output resonator coupled to said beam to take amplified output from the electrons thereof after they have passed and been modulated by the input resonator, said input and output resonators being resonant at substantially equal frequencies, means for taking output signals directly from said output resonator, and means for applying feedback reaction, insufficient to cause self-oscillations, over at least one intermediate length of said modulated beam which is beyond said input resonator in the direction of electron flight, said intermediate length being a length over which said modulated beam remains modulated in accordance with said input signals, wherein the feedback reaction means comprises at least one pair of additional resonators situated between the input resonator and the output resonator and coupled to the electrons of the beam at the opposite ends of an intermediate length thereof, and a feedback path so dimensioned as to provide positive energy feedback between that additional resonator which is nearer the output resonator and the other additional resonator.

Requested 1-27-64

2,960,659

SEMICONDUCTIVE ELECTRON SOURCE

Joseph A. Burton, Chatham, N.J., assignor to Bell Telephone Laboratories, Incorporated, New York, N.Y., a corporation of New York

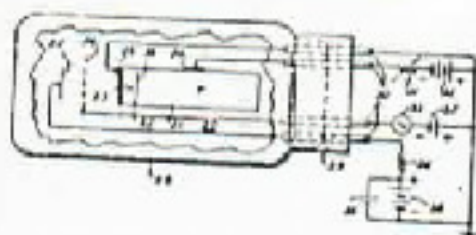
Filed Sept. 1, 1955, Ser. No. 532,043

13 Claims. (Cl. 330-65)

2. An electron source comprising a semiconductive body the main portion of which is of p conductivity type and an end surface layer of which is of n conductivity type, means for biasing the p-n junction between said main portion and said surface layer in the reverse direc-

Pat. 2,960,659

beyond the point of avalanche breakdown, and a coating of electron work function reducing material on the surface of said surface layer for inducing electron emission from said surface into free space.

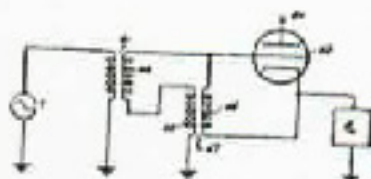


2,960,660

FEEDBACK AMPLIFIER CIRCUIT

Frank Stuart Boxall, Palo Alto, Calif., assignor to Lenkurt Electric Co., Inc., San Carlos, Calif., a corporation of Delaware

Filed June 7, 1957, Ser. No. 664,221
1 Claim. (Cl. 330-91)



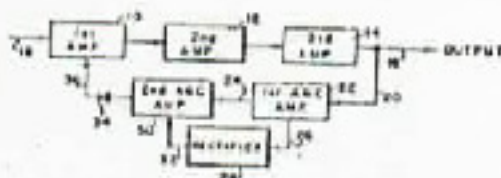
In a power amplifier producing output signals of lesser amplitude than the amplitude of input signals and including a multi-element vacuum tube connected in cathode-follower arrangement to a load circuit, an input transformer having a primary winding connected to receive signals and a secondary winding connected at one end thereof to a control electrode of said vacuum tube, and a control transformer of substantially one-to-one transformation and having a first winding connected between the control electrode and cathode of said vacuum tube and a second winding connected in series between the secondary winding of said input transformer and ground for adding to input signals thereat a control signal proportional to the difference between input and output of said vacuum tube.

2,960,661

AUTOMATIC GAIN CONTROL CIRCUIT

Raymond W. Bratschi, St. Paul, Minn., assignor to Sperry Rand Corporation, New York, N.Y., a corporation of Delaware

Filed Jan. 16, 1957, Ser. No. 634,447
11 Claims. (Cl. 330-127)



1. An automatic gain control circuit comprising a gain amplification channel including at least a first amplifier whose gain is to be controlled, amplification means, means for applying a signal as it appears in said channel to said amplification means for amplification thereof, means for applying the output of said amplification means as a negative control bias to said first amplifier to cause the output of said channel to be relatively constant in amplitude, and means for increasing the control of the channel output regardless of amplitude variations, at least above a relatively low amplitude, in said signal as received by said channel comprising means for deriving from said signal as it appears in said chan-

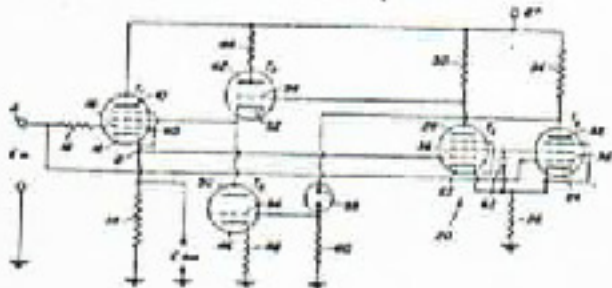
nel a direct current bias having an amplitude varying directly as the amplitude of said signal, means applying said direct current bias to said amplification means for increasing the gain of the said amplification means when the amplitude of said direct current bias increases to make the channel output amplitude even more constant.

2,960,662

WIDE RANGE SIGNAL LEVEL TRANSFER CIRCUIT

Gordon E. Nelson, Windsor, Conn., assignor to Combustion Engineering, Inc., New York, N.Y., a corporation of Delaware

Filed July 8, 1958, Ser. No. 747,237
6 Claims. (Cl. 330-128)



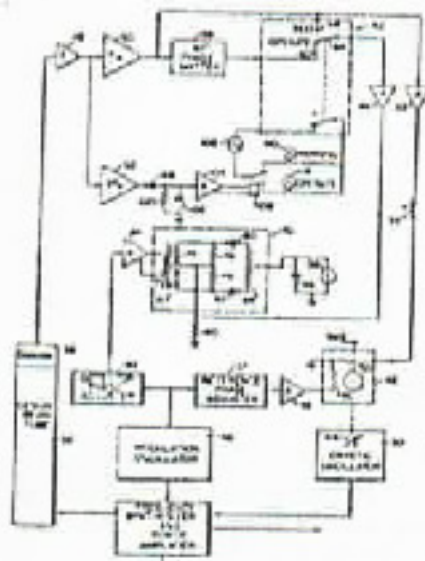
1. In an organization of the type described a cathode follower tube having a supplemental grid intermediate the control grid and the plate, means responsive to a difference in potential between the control voltage and the cathode effective to adjust the voltage of said supplemental grid in a manner to change the cathode current of the cathode follower so as to bring the cathode potential and the control voltage into substantial agreement, said means including a difference amplifier connected to amplify any difference between the control voltage and the voltage of the cathode and means responsive to the output of the voltage difference amplifier operative to produce a voltage that varies in sense in accordance therewith and which is applied to said supplemental grid.

2,960,663

FREQUENCY CONTROL APPARATUS

Walter A. Mainberger, Winchester, Mass., assignor to National Company, Inc., Malden, Mass., a corporation of Massachusetts

Filed June 26, 1958, Ser. No. 744,729
12 Claims. (Cl. 331-3)



1. A frequency control system comprising, in combination, a frequency standard, frequency controllable means for generating a signal to be applied to said standard, a modulation signal generator, means for fre-

2,082,161

DEHYDRATION OF ANTIMONY TRIFLUORIDE AND MANUFACTURE OF FLUORINATED ALIPHATIC COMPOUNDS

Albert L. Henne, Columbus, Ohio, assignor, by mesne assignments, to General Motors Corporation, a corporation of Delaware

No Drawing. Application January 30, 1931, Serial No. 512,469. Renewed October 16, 1935

1 Claim. (Cl. 260-162)

The manufacture of halo-fluoro aliphatic derivatives of hydrocarbons which comprises mixing wet antimony trifluoride and an azeotropic distillation agent, evaporating the distillation agent and water from the fluoride, reacting the antimony fluoride thus obtained with a halogen derivative of an aliphatic hydrocarbon, containing a halogen other than fluorine.

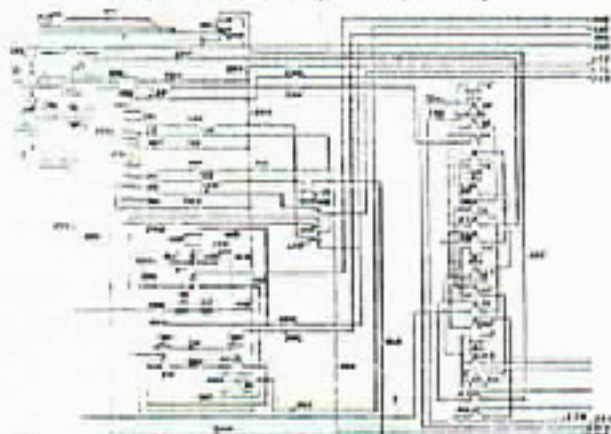
2,082,162

TRAIN DISPATCHING SYSTEM FOR RAILROADS

Forest B. Hitchcock and Oscar H. Dicke, Rochester, N. Y., assignors to General Railway Signal Company, Rochester, N. Y.

Application March 9, 1929, Serial No. 345,667

24 Claims. (Cl. 246-3)



In a train dispatching system of the wayside control type, the combination with a large number of wayside signals located along the railway track and each controlled by traffic conditions ahead, of means for controlling said signals selectively over the same line wires comprising, a bank of relays in the dispatcher's office, a bank of relays at each of a plurality of way stations, means for operating the relays of each bank sequentially in a particular order, and over an uninterrupted line circuit, in such a way that corresponding relays of all the banks are operated synchronously and so that corresponding relays of all banks at all times assume corresponding positions, and a plurality of message circuits for controlling said signals closed sequentially and each including the same line wire and including contacts of corresponding relays of the dispatcher's office bank and a particular way station bank.

2,082,163

SOIL EROSION CONTROLLING APPARATUS

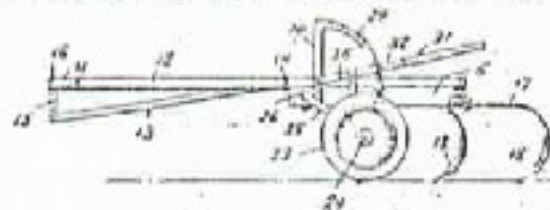
Roy Jeffroy, Perryton, Tex.

Application June 18, 1936, Serial No. 86,015

1 Claim. (Cl. 55-74)

In a ground working machine, a wheeled frame, a draft appliance attached to the frame, said draft appliance comprising a pair of angle bars each secured at its rear end portion to the frame and extending forwardly from the frame in close spaced parallel relation to each other, a hitch member having its upper end secured to the forward end of the angle bars and between the same,

and a brace bar extension rearwardly from the lower end of the hitch member and having its



rear end secured to the said angle bars at a point between the intermediate portions thereof.

2,082,164

TUBE FITTING

Sebastian Karrer, Washington, D. C., assignor to Consolidated Gas Electric Light and Power Company of Baltimore, Baltimore, Md., a corporation of Maryland

Application March 13, 1934, Serial No. 715,367

4 Claims. (Cl. 285-86)



1. In a coupling for a thin-walled tube to be subjected to freezing and thawing temperatures and wherein the coupling nut clamps the end of the tube against a fitting member and surrounds the tube with a relatively slight clearance in which moisture may collect and freeze, said coupling nut having means for relieving pressure due to the freezing of moisture in the clearance between the nut and tube, said means including one or more expansion recesses leading from said clearance outwardly to the atmosphere.

2,082,165

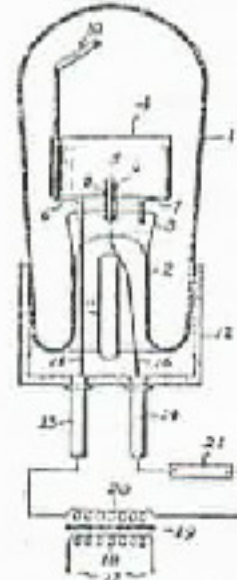
GASEOUS DISCHARGE RECTIFIER

James D. Le Van, Belmont, Mass., assignor to Raytheon Production Corporation, Newton, Mass., a corporation of Delaware

Application October 8, 1932, Serial No. 636,852

Renewed October 30, 1936

8 Claims. (Cl. 250-27.5)



1. A gaseous discharge device comprising a hermetically-sealed envelope containing an ionizable gas, a glow discharge cathode of extended area and an anode, said electrodes being adapted to support an ionizing discharge through said